# Diameter-7 cells from a spread and its whiskers: the freed sets and the bounds for every q

Degree–diameter lab, 29 October 2026 (job thm-conics)

*Abstract.* The jobs "conics" and "conics-2" raised the lab's values at the diameter-7 cells $(14,7)$ to $(20,7)$ by freeing, on the friendly transitions of the typed hexagon layouts, two sets: the points of the Hermitian spread $S$ of $H(q)$, and the lines meeting a spread line $v$ together with an exact whisker set and the $q^2$ spread lines that carry no whisker. Here these steps are turned into statements for every prime power $q$. (1) For any generalized hexagon of order $(q,q)$, any spread $S$, any $v \in S$ and any whisker set $W$, the lines meeting $v$, the whiskers and the spread lines missed by the whiskers form a dual segment set; for the exact whisker set of `POLISHED.md` (every odd $q$ and every even $q \geq 4$) it has exactly $q^4+q^2+q$ lines and is maximal, while the spread points form a segment set of $(q+1)(q^3+1)$ points. (2) For $2k+1$ levels, any regular tournament and any typing, some level has at most $\lfloor k/2 \rfloor$ friendly outgoing transitions and some level at most $\lfloor k/2 \rfloor$ friendly incoming ones; this bounds the hub counts, with the exact second-order term, and the alternating typing attains the bound for the total number of hubs for every $k$. (3) With the $q^2+1$ points of an elliptic quadric section on the non-friendly transitions, this gives, for every $k \geq 2$ and every prime power $q \geq q_k$,

$$n_b(q+1+k,7) \;\geq\; 2(2k+1)(q+1)(q^4+q^2+1) + \begin{cases} k\,(q^4+2q^2+q+1) & k \text{ even}, \\ \frac{k-1}{2}\,(2q^4+q^3+q^2+2q+1)+(k+1)(q^2+1) & k \text{ odd}, \end{cases}$$

with $q_2 = 16$, $q_3 = q_4 = 23$, $q_5 = 29$, $q_k \leq 300k$ in general. The hub graphs are given by a new explicit construction (sums of blown-up hexagons); smaller $q$ are checked by computer where possible. The formulas reproduce the vertex counts of all fifteen graphs built in the two jobs. At $(19,7)$ the formula equals the built value 11,316,940 and every ingredient is proved. At $(18,7)$ the theorem with an explicit hub graph (two copies of the incidence graph of $H(7)$ joined by perfect matchings) gives 7,318,644, above the lab's value 7,309,092; that graph was built, and checked from every hub (§5.1).

*Labels.* Every result is marked **Theorem** (proved here or in the cited note), **Computer-checked** (a program checked it; the program is in the repository) or **Heuristic**, and **[new]** or **[restatement]**. "New" means new in the lab's notes; the literature check of `D7_HEXAGON.md` §11 applies, and we did not search the literature again.

*Sources.* `research/synthetic/CONICS.md` §3.4 (branch `claude/conics-generalized-polygons-6gjqxi`); `research/synthetic/conics2/README.md` (branch `claude/conics-2-diameter-7-ic0oat`, state of 11:44 UTC); `research/theorems/D7_HEXAGON.md` (branch `claude/d7-hexagon-theorems-4w1prt`), whose Definitions 2.1, 3.1, Theorems A, B, C, 3.4 and Proposition 5.2 are used throughout; `research/theorems/POLISHED.md` §6 (branch `claude/thm-polish-2oddll`); `research/synthetic/SYNTHETIC.md` (branch `claude/synthetic-incidence-geometry-s4m08f`), whose conventions we follow. Programs of this note: `research/synthetic/d7thm/`; squad round `research/squads/d7thm-r1/`.

## 0. Setting

**Hexagons.** A generalized hexagon $\Gamma$ of order $(q,q)$ has $N = N(q) = (q+1)(q^4+q^2+1)$ points and as many lines; $q+1$ points on a line and $q+1$ lines on a point. Distances are taken in the incidence graph $I$ (diameter 6, girth 12); elements at distance 6 are *opposite*. For an element $x$, $|\Gamma_i(x)| = (q+1)q^{i-1}$ for $1 \leq i \leq 5$. A *spread* is a set of $q^3+1$ pairwise opposite lines; every line not in a spread $S$ meets exactly one line of $S$, in one point (`D7_HEXAGON.md`, Lemma 1.2(c)). The example throughout is the split Cayley hexagon $H(q)$ in $Q(6,q)$ and its Hermitian spread: the lines of $H(q)$ inside a hyperplane $\Pi$ with $\Pi \cap Q(6,q)$ elliptic (every $q$; [Th80]).

**Segment sets** (D7_HEXAGON.md, Definition 1.3). A point set $F$ is a *segment set* if for every $p \in F$ the points of $F \setminus \{p\}$ collinear with $p$ lie on one line through $p$. A line set $G$ is a *dual segment set* if for every $\ell \in G$ the lines of $G \setminus \{\ell\}$ that meet $\ell$ pass through one point of $\ell$. Both kinds have at most $(q+1)(q^3+1)$ elements (D7_HEXAGON.md, Proposition 1.4).

**Layouts.** $L = 2k+1$ *levels*, each a copy of $I$ of *type* $H$ (colour $\mathcal{A}$ = points, $\mathcal{B}$ = lines) or $D$ (colour $\mathcal{A}$ = lines, $\mathcal{B}$ = points). The levels are the vertices of a *regular tournament* $T$: every two levels are joined by exactly one *transition* $j \to j'$, and every level has $k$ outgoing and $k$ incoming transitions. A transition is a bijection from the $\mathcal{A}$-elements of level $j$ to the $\mathcal{B}$-elements of level $j'$; it is *friendly* if the types differ, and then it is the identity. D7_HEXAGON.md uses the rotational tournament $j \to j+i$ ($1 \le i \le k$, indices mod $L$); Theorem A and Theorem 3.4 there hold verbatim for every regular tournament, since their proofs only use that every pair of levels is the pair of one transition. The *freed set* $R_t$ of a transition is the set of $\mathcal{A}$-elements whose matching edge is deleted. Level $j$ then has $\alpha_j = \sum_{t \text{ out of } j} |R_t|$ free $\mathcal{A}$-slots and $\beta_j = \sum_{t \text{ into } j} |R_t|$ free $\mathcal{B}$-slots. $K_{\mathcal{B}} \le \min_j \alpha_j$ hubs of colour $\mathcal{B}$ each take one free $\mathcal{A}$-slot in every level, $K_{\mathcal{A}} \le \min_j \beta_j$ hubs of colour $\mathcal{A}$ one free $\mathcal{B}$-slot in every level, and a bipartite *hub graph* $Y$ joins the two kinds of hubs.

**What repairs a transition.** Theorem B of D7_HEXAGON.md: a friendly transition is repaired if its freed set is a segment set (source of type $H$) or a dual segment set (source of type $D$). Theorem C there, with Corollary 5.3(a): a non-friendly transition $H \to H$ with matching $\sigma$ is repaired by a set $R$ of $m$ pairwise opposite points when $\sigma$ maps $R$ onto $m$ pairwise opposite lines and the points collinear with $x$ onto the lines concurrent with $\sigma(x)$ for every $x \in R$; such a $\sigma$ exists for every partial ovoid $R$ and every partial spread of the same size. A $D \to D$ transition uses $\sigma^{-1}$ and frees $\sigma(R)$.

**Hub graphs.** $\mathrm{HG}(K_1, K_2; \delta)$ denotes a bipartite graph with colour classes of sizes $K_1$ and $K_2$, maximum degree at most $\delta$, in which any two vertices *of the same colour* are at distance at most 6.

**Theorem 0.1** (D7_HEXAGON.md, Theorem 3.4) [restatement]. If every transition is repaired, $K_{\mathcal{B}} \le \min \alpha_j$, $K_{\mathcal{A}} \le \min \beta_j$ and the hub graph is an $\mathrm{HG}(K_{\mathcal{B}}, K_{\mathcal{A}}; q-k)$, then the graph has maximum degree $q+1+k$ and diameter at most 7, and

$$n_b(q+1+k, 7) \;\ge\; 2(2k+1)N(q) + K_{\mathcal{A}} + K_{\mathcal{B}}.$$

The value $2(2k+1)N(q)$ alone (no freed sets, no hubs) is the value of the 2013 table at the cells $(7,7)$, $(9,7)$–$(16,7)$ [MS13], and the lab's beyond-table value at $(17,7)$–$(20,7)$.

**What uses coordinates.** Sections 2 and 4 are about tournaments and bipartite graphs only. In §1, Theorem 1.4, Proposition 1.5 and Lemma 1.6 hold in every generalized hexagon of order $(q,q)$ with a spread; only Theorem 1.7 (the whisker set of POLISHED.md meets each spread line in 0 or $q+1$ points, so exactly $q^2$ spread lines are missed) and the existence statements (Hermitian spread, exact whisker set, elliptic quadric section) use the coordinates of $H(q)$. Coordinates appear only in proofs.

**Notation for sizes.** $P(q) = (q+1)(q^3+1) = q^4+q^3+q+1$, $\Lambda(q) = q^4+q^2+q$, and $m$ the size of the non-friendly sets. Note $P - \Lambda = q^3 - q^2 + 1$.

# 1. The freed sets

## 1.1 The configuration

**Definition 1.1.** Let $S$ be a spread of $\Gamma$, $\mathcal{P}_S$ the set of points on its lines, and $v \in S$. $\mathcal{L}_v$ is the set of the $q(q+1)$ lines other than $v$ that meet $v$. A point is *free* if it is at distance 5 from $v$ and not in $\mathcal{P}_S$. A *whisker set* is a set $W$ of lines, each opposite $v$ and not in $S$, such that no free point lies on two lines of $W$; it is *exact* if every free point lies on exactly one line of $W$. Finally

$$S_0(W) = \{n \in S \setminus \{v\} : \; n \text{ meets no line of } W\}, \qquad R_L(W) = \mathcal{L}_v \cup W \cup S_0(W).$$

(This is the configuration of Theorem D, POLISHED.md §6.2 and SYNTHETIC.md §3.2, written without coordinates.)

**Lemma 1.2** [restatement, with the counts; Theorem]. (a) The points at distance at most 3 from $v$ are the $q+1$ points of $v$ and the $q^2(q+1)$ points at distance 3, each of which lies on exactly one line of $\mathcal{L}_v$. (b) The points of $\mathcal{P}_S$ not on $v$ are at distance 5 from $v$. (c) There are $q^3(q^2-1)$ free points. (d) A line opposite $v$ and not in

$S$ has exactly one point in $\mathcal{P}_S$ and $q$ free points. (e) A whisker set has at most $q^2(q^2-1)$ lines, with equality if and only if it is exact.

*Proof.* (a) A point at distance 3 from $v$ lies on a line meeting $v$; two such lines through it would close a circuit of length 4 or 6. Counting: $|\Gamma_3(v)| = (q+1)q^2$. (b) A point $x$ on a spread line $n \neq v$ has $d(x,v)$ odd, at most 5 and at least $d(n,v)-1 = 5$. (c) $|\Gamma_5(v)| = (q+1)q^4$, of which $q^3(q+1)$ lie in $\mathcal{P}_S$ by (b). (d) Every point of a line opposite $v$ is at distance 5 from $v$; the line meets exactly one line of $S$, in one point. (e) By (d) and (c), counting incidences between free points and $W$. □

## 1.2 Spread points and whiskers

**Proposition 1.3** (`D7_HEXAGON.md`, Proposition 1.4) [restatement; Theorem]. $\mathcal{P}_S$ is a segment set of $P(q) = (q+1)(q^3+1)$ points, and no segment set is larger; equality holds only for the point sets of spreads.

**Theorem 1.4** [new in this form; Theorem]. Let $S$ be a spread, $v \in S$ and $W$ any whisker set.

(a) $R_L(W) = \mathcal{L}_v \cup W \cup S_0(W)$ is a dual segment set, and two of its lines that meet, meet in a point of $\mathcal{P}_S$.

(b) $|R_L(W)| = q(q+1) + |W| + |S_0(W)|$.

(c) If $W$ is exact and every point of $\mathcal{P}_S$ not on $v$ lies on a line of $W$ or of $S_0(W)$, then every point of $\Gamma$ lies on a line of $R_L(W)$, and no line can be added to $R_L(W)$: it is a maximal dual segment set.

(d) A line $n \in S \setminus \{v\}$ all of whose whiskers pass through one point of $n$ can be added to $R_L(W)$ as well.

*Proof.* (a) A line of $\mathcal{L}_v$ has one point on $v \subseteq \mathcal{P}_S$ and its other $q$ points at distance 3 from $v$, so not in $\mathcal{P}_S$ (Lemma 1.2(b)). A whisker has one point in $\mathcal{P}_S$ and $q$ free points (Lemma 1.2(d)). So every line of $\mathcal{L}_v \cup W$ has exactly one point in $\mathcal{P}_S$. Its other points are either at distance 3 from $v$, and then on no other line of $\mathcal{L}_v$ (Lemma 1.2(a)) and on no whisker (whose points are at distance 5), or free, and then on no line of $\mathcal{L}_v$ and on at most one whisker. So two lines of $\mathcal{L}_v \cup W$ never meet outside $\mathcal{P}_S$. A line $n \in S_0(W)$ meets no line of $\mathcal{L}_v$ (whose points are at distance at most 3 from $v$, while those of $n$ are at distance 5), no whisker (definition) and no other spread line. So the lines of $S_0(W)$ are isolated, and for every other line $\ell$ of $R_L(W)$ the lines of $R_L(W)$ meeting $\ell$ all pass through the unique point of $\ell$ in $\mathcal{P}_S$. (b) The three sets are disjoint: whiskers are opposite $v$, lines of $\mathcal{L}_v$ are not, and neither kind is in $S$. (c) Points of $v$ and points at distance 3 from $v$ lie on lines of $\mathcal{L}_v$, free points on whiskers ($W$ exact), and the points of $\mathcal{P}_S$ off $v$ by hypothesis. A line $\ell \notin R_L(W)$ then meets a line of $R_L(W)$ at each of its $q+1$ points, and these lines are different from each other (two points of $\ell$ on one line $\neq \ell$ is impossible), so $R_L(W) \cup \{\ell\}$ is not a dual segment set. (d) Such an $n$ meets lines of $R_L(W)$ only in that point $x$ (it meets no line of $\mathcal{L}_v$ and no other spread line), and for a whisker through $x$ the new line meets it at its point in $\mathcal{P}_S$, where all its other meetings already are. Two such spread lines do not meet. □

For $W$ exact and $S_0(W)$ left out this is `CONICS.md`, Lemma 3.1; for the whisker set of `POLISHED.md` and its $q^2$ empty spread lines it is Proposition 1(c),(d) of `conics2/README.md`. The proof above uses only incidences.

**Proposition 1.5** [new; Theorem] (*where the room is*). Let $G$ be a dual segment set in which any two lines that meet do so in a point of $\mathcal{P}_S$ (as in Theorem 1.4(a)). Then $|G \setminus S| \leq q^4+q$, with equality if and only if the lines of $G \setminus S$ cover every point outside $\mathcal{P}_S$ exactly once. Hence $|G| \leq q^4+q+|G \cap S|$. For an exact whisker set, $\mathcal{L}_v \cup W$ has exactly $q^4+q$ lines.

*Proof.* A line not in $S$ meets exactly one spread line, in one point, so it has exactly one point in $\mathcal{P}_S$ (a second point would lie on a second spread line). Its other $q$ points lie outside $\mathcal{P}_S$, and by hypothesis these $q$-sets are disjoint for different lines of $G \setminus S$. There are $N - P(q) = q^5 + q^2 = q(q^4+q)$ points outside $\mathcal{P}_S$. Finally $q(q+1) + q^2(q^2-1) = q^4+q$. □

So within this class $R_L(W)$ can only grow by containing more spread lines, at most $q^3-q^2+1$ more; the general bound $(q+1)(q^3+1) = q^4+q+(q^3+1)$ is the case where all spread lines are in $G$, which forces an ovoid.

**Lemma 1.6** [new; Theorem]. Let $W$ be exact. Every whisker meets exactly one line of $S \setminus \{v\}$, and a line $n \in S \setminus \{v\}$ meets at most $q^2-1$ whiskers. So at least $q^2$ spread lines other than $v$ meet a whisker and $|S_0(W)| \leq q^3-q^2$. If every line of $S \setminus \{v\}$ meets either no whisker or $q+1$ whiskers, then $|S_0(W)| = q^2$.

*Proof.* A whisker meets one line of $S$ (§0), not $v$, since it is opposite $v$. At a point $x$ of $n$ the $q+1$ lines are $n$, the line through $x$ on the geodesic to $v$ (at distance 4 from $v$, so not a whisker) and $q-1$ others. Hence $n$ meets

at most $(q+1)(q-1)$ whiskers, and $q^2(q^2-1)/(q^2-1) = q^2$. In the last case the lines meeting whiskers number $q^2(q^2-1)/(q+1) = q^2(q-1)$, and $q^3 - q^2(q-1) = q^2$. □

## 1.3 The exact whisker set of `POLISHED.md`

In $H(q)$ with the coordinates of `POLISHED.md` §6.1 ($q$ odd) and §6.5 ($q$ even): $\Pi$ the elliptic hyperplane $X_0 + cX_4 = 0$ ($q$ odd, $-c$ a non-square) or $X_3 = X_0 + \delta X_4$ ($q$ even, $\operatorname{Tr}\delta = 1$), $S$ its Hermitian spread, $v = \langle e_1, e_6\rangle$, the hyperplanes $\Pi_z \supset v$ ($z = (\alpha, \beta) \in \mathbb{F}^2$) and their spreads $S_z$, the function $\nu$ on the points of $\Pi$ off $v$, the zero-free cubic $G$ and $F(z) = G(z)/N(z)$ with $N$ the anisotropic norm form; $W = \{\ell \in S_z \ \ \{v\} : z \neq 0,\ \nu(\ell \cap \Pi) = F(z)\}$ is exact for every odd $q$ (Theorem 6.5) and every even $q \geq 4$ (Theorem 6.14).

**Theorem 1.7** (`conics2/README.md`, Proposition 1(a),(b)) [restatement with proof; Theorem]. Let $q$ be odd, or even with $q \geq 4$, and $W$ the exact whisker set above. Then every point $X$ of $\mathcal{P}_S$ not on $v$ lies on exactly one whisker if $\nu(X) \neq 0$ and on none if $\nu(X) = 0$; $\nu$ is constant on every line of $S \ \ \{v\}$. Hence every line of $S \ \ \{v\}$ meets either $q+1$ whiskers, one at each of its points, or none, and $S_0(W) = \{n \in S \ \ \{v\} : \nu(n) = 0\}$ has exactly $q^2$ lines.

*Proof.* Whiskers through $X$ are whiskers whose point in $\Pi$ is $X$ (their other points are off $\Pi$). On $\Pi$ the equation of $\Pi_z$ reduces to $\alpha X_2 + \beta X_5 = 0$ (both parities), and $(X_2, X_5) \neq (0,0)$ because $X$ is not on $v$ (`POLISHED.md`, Lemma 6.1(d) / 6.10). So the $z \neq 0$ with $X \in \Pi_z$ form the punctured line $\{tz_0 : t \neq 0\}$, $z_0 = (X_5, -X_2)$. For each of them exactly one line of $S_z$ passes through $X$ (Lemma 6.1(b); for even $q$ read Lemma 6.10 for 6.1 throughout); it is not $v$, it is not in $S$ (6.1(e)), so it meets $\Pi$ only in $X$ (6.1(c)), and different $z$ give different lines (6.1(e)). It is a whisker if and only if $\nu(X) = F(tz_0)$. As $G$ is a cubic form and $N$ a quadratic form, $F(tz_0) = tF(z_0)$, and $F(z_0) \neq 0$ since $G$ has no zero. So $t \mapsto F(tz_0)$ is a bijection of $\mathbb{F}^\times$: one whisker through $X$ if $\nu(X) \neq 0$, none if $\nu(X) = 0$. $\nu$ is constant on spread lines by `POLISHED.md`, Lemma 6.6(d) / 6.12(d). The count is Lemma 1.6. □

**Corollary 1.8** [Theorem]. For every prime power $q$ with $q$ odd or $q \geq 4$, in $H(q)$: $R_H = \mathcal{P}_S$ is a segment set with $P(q) = q^4 + q^3 + q + 1$ points, and $R_L = \mathcal{L}_v \cup W \cup S_0(W)$ is a maximal dual segment set with $\Lambda(q) = q^4 + q^2 + q$ lines (Theorem 1.4 with Theorem 1.7). In the notation of Proposition 1.5, $\mathcal{L}_v \cup W$ is extremal and $|R_L \cap S| = q^2$.

*Computer-checked.* `conics2/typed_build.py` asserts every statement of Theorem 1.7 and Corollary 1.8 for $q = 4, 5, 7, 11, 13, 16$ inside the builds. Independently, squad round `d7thm-r1` (own code, not the builder): card c1 for $q = 3, 5, 7$ (hexagon built from the quadric and Tits' equations), card c2 for $q = 9, 11$, card c3 for $q = 2, 4, 8$: exactness of $W$, $\nu$ constant on spread lines, 0 or $q+1$ whiskers per spread line, $|S_0| = q^2$ equal to the lines with $\nu = 0$, the dual segment property, $|R_L| = q^4 + q^2 + q$ (6,651 at $q = 9$, 14,773 at $q = 11$, 276 at $q = 4$, 4,168 at $q = 8$), maximality (every outside line tested), and the segment property of $\mathcal{P}_S$; card c2 also for every admissible $c$ at $q = 9, 11$, card c3 for every admissible $(c_0, c_1)$ at $q = 4, 8$; negative controls fail as they should (`research/squads/d7thm-r1/scoreboard.jsonl`).

*Remarks.* (i) $q = 2$: Lemma 6.11 of `POLISHED.md` gives no zero-free $G$ (card c3 re-checked all four pairs), but an exact whisker set of 12 lines exists (`POLISHED.md`, Corollary 6.15), so Theorem 1.4 gives $|R_L| \geq 18$; the largest dual segment set of $H(2)$ has 25 lines (**Computer-checked**, `conics2/README.md` §3: SAT and MaxSAT, squad round `conics2-r1` cards c6, c7). (ii) $R_L$ is not maximum in general: $H(4)$ has a dual segment set of 288 $>$ 276 lines (**Computer-checked**, same place: card c8, re-checked by `research/squads/conics2-r1/tools/dualseg_check.py`; it is not of the kind in Proposition 1.5). Whether $q^4 + q^2 + q$ can be beaten for $q \geq 5$ is open; by Proposition 1.5 a set of this kind would need more than $q^2$ spread lines. (iii) For $q = 3^h$ the pencils of an ovoid give a dual segment set of $P(q)$ lines, and the duality layout of `D7_HEXAGON.md`, Corollary 8.2 (every freed set of size $P(q)$) is at least as good as any typed layout, by Theorem 2.1. (iv) We know no coordinate-free description of the $q^2$ lines with $\nu = 0$.

## 1.4 The non-friendly sets

**Proposition 1.9** (`D7_HEXAGON.md`, Lemma 1.2(a), Corollaries 5.3(a) and 9.1) [restatement; Theorem]. For every prime power $q$, the $q^2 + 1$ points of an elliptic quadric section $Q^-(3, q)$ of $Q(6, q)$ are pairwise opposite in $H(q)$, and with any $q^2 + 1$ lines of the Hermitian spread they give a local duality $\sigma$ as in Theorem C. So $m = q^2 + 1$ is available on every non-friendly transition.

The lab's builds use larger non-friendly sets found by computer at $q = 11$ ($m = 768$, mirrored set) and $q = 13$ ($m = 792$), and $m = q^2 + 1 = 257$ at $q = 16$ (**Computer-checked**: the builder re-checks non-collinearity,

non-concurrency and the local duality at every point).

## 2. Slot counts

Let $T$ be a regular tournament on $2k+1$ levels with a typing. For a level $j$ let $f_j$ be the number of friendly transitions out of $j$ and $g_j$ the number into $j$. With friendly sets of sizes $P$ (points, source $H$) and $\Lambda$ (lines, source $D$) and non-friendly sets of size $m$,

$$\alpha_j = f_j X_j + (k-f_j)m, \quad X_j = \begin{cases} P & \tau_j = H \\ \Lambda & \tau_j = D \end{cases}, \qquad \beta_j = g_j Y_j + (k-g_j)m, \quad Y_j = \begin{cases} \Lambda & \tau_j = H \\ P & \tau_j = D \end{cases}$$

(a friendly transition into an $H$ level comes from a $D$ level, which frees lines). Put $a = \lfloor k/2 \rfloor$ and assume

$$P \;\geq\; \Lambda \;\geq\; m, \qquad \Lambda - m \;\geq\; a\,(P-\Lambda). \tag{$*$}$$

For the sets of §1, $(*)$ holds whenever $k \leq 2q+1$ and $m \leq q^2+1$, since then $a(q^3-q^2+1) \leq q^4-q^3+q \leq q^4+q-1$; at $q=11$, $m=768$ it holds for $k \leq 23$, at $q=13$, $m=792$ for $k \leq 27$. The bound needs $k \leq q-1$ anyway (hub degree $q-k \geq 1$).

**Theorem 2.1** [new; Theorem]. For every regular tournament on $2k+1$ levels and every typing, with $a = \lfloor k/2 \rfloor$:

(a) $\min_j f_j \leq a$ and $\min_j g_j \leq a$;

(b) if $k$ is even, it is impossible that $f_j \geq a$ at every level and $f_j \geq a+1$ at every $D$ level; likewise with $g$ and $H$ levels;

(c) if $k$ is odd, it is impossible that $f_j \geq a+1$ at every $D$ level and $g_j \geq a+1$ at every $H$ level.

Consequently, if $P \geq \Lambda \geq m$: for $k$ odd, $\min\alpha, \min\beta \leq aP+(a+1)m$ and $\min\alpha + \min\beta \leq a(P+\Lambda)+2(a+1)m$; for $k$ even and under $(*)$, $\min\alpha \leq a(\Lambda+m)$ and $\min\beta \leq a(\Lambda+m)$.

*Proof.* Let $h$ and $d$ be the numbers of $H$ and $D$ levels. The $k-f_j$ non-friendly transitions out of an $H$ level go to $H$ levels, so $\sum_{\tau_j=H}(k-f_j) = \binom{h}{2}$, and likewise $\sum_{\tau_j=D}(k-f_j) = \binom{d}{2}$ and the same with $g$.

(a) If $f_j \geq a+1$ everywhere, then $\binom{h}{2} \leq h(k-a-1)$ and $\binom{d}{2} \leq d(k-a-1)$, so $h+d \leq 4k-4a-2 \leq 2k$, less than $2k+1$. For $g$ the same.

(b) $k=2a$. From $f_j \geq a$ on $H$ levels, $\binom{h}{2} \leq ha$, so $h \leq 2a+1$; from $f_j \geq a+1$ on $D$ levels, $\binom{d}{2} \leq d(a-1)$, so $d \leq 2a-1$. Then $h+d \leq 4a < 4a+1$. For $g$ exchange the roles of $H$ and $D$.

(c) $k=2a+1$. On $H$ levels $g_j \geq a+1$ gives $\binom{h}{2} \leq ha$ (count non-friendly incoming transitions), so $h \leq 2a+1$; on $D$ levels $f_j \geq a+1$ gives $\binom{d}{2} \leq da$, so $d \leq 2a+1$. Then $h+d \leq 4a+2 < 4a+3$.

Consequences. A level with $f_j \leq a-1$ has $\alpha_j \leq (a-1)P+(a+1)m \leq a\Lambda+am$, by $(*)$ (only $\Lambda-m \geq (a-1)(P-\Lambda)$ is used, which always holds for $k=2$). If $k$ is even and $\min\alpha > a(\Lambda+m)$, every level has $f_j \geq a$, and every $D$ level $f_j \geq a+1$ (a $D$ level with $f_j = a$ has $\alpha_j = a\Lambda+am$), contradicting (b). If $k$ is odd, a level with $f_j \leq a$ has $\alpha_j \leq aP+(a+1)m$, so by (a) $\min\alpha \leq aP+(a+1)m$, and the same for $\beta$. If $\min\alpha+\min\beta > a(P+\Lambda)+2(a+1)m$, both minima exceed $a\Lambda+(a+1)m$. A $D$ level with $f_j \leq a$ has $\alpha_j \leq a\Lambda+(a+1)m$ and an $H$ level with $g_j \leq a$ has $\beta_j \leq a\Lambda+(a+1)m$, so every $D$ level has $f_j \geq a+1$ and every $H$ level $g_j \geq a+1$, contradicting (c). □

**Theorem 2.2** [new; Theorem] (*the alternating layout*). Take the rotational tournament $j \to j+i$ $(1 \leq i \leq k)$ on $\mathbb{Z}_{2k+1}$ and the typing $\tau_j = H$ for $j$ even, $\tau_j = D$ for $j$ odd $(0 \leq j \leq 2k$; the levels $2k$ and $0$ are both of type $H)$. Then $f_j = \lceil k/2 \rceil$ for $0 \leq j \leq k$, and for $j = k+t$ $(1 \leq t \leq k)$

$$f_j = \begin{cases} k/2 \text{ ($t$ even, $H$)} \quad k/2+1 \text{ ($t$ odd, $D$)} & k \text{ even}, \\ (k-1)/2 \text{ ($t$ odd, $H$)} \quad (k+1)/2 \text{ ($t$ even, $D$)} & k \text{ odd}; \end{cases}$$

and $g_j = f_{2k-j}$. Hence (for odd $k$ using $(*)$)

$$k \text{ even:} \quad \min\alpha = \min\beta = \tfrac{k}{2}(\Lambda+m); \qquad k \text{ odd:} \quad \min\alpha = \tfrac{k-1}{2}P + \tfrac{k+1}{2}m, \quad \min\beta = \tfrac{k-1}{2}\Lambda + \tfrac{k+1}{2}m,$$

For $k$ even both minima attain the bounds of Theorem 2.1; for $k$ odd $\min\alpha$ and $\min\alpha+\min\beta$ do.

*Proof.* For $j\le k$ the targets $j+1,\dots,j+k$ need no reduction mod $2k+1$, and exactly $\lceil k/2\rceil$ of them have the parity opposite to $j$. For $j=k+t$ the targets are $j+1,\dots,2k$ ($k-t$ of them, $\lceil(k-t)/2\rceil$ of the other parity) and $0,\dots,t-1$, of which $\lfloor t/2\rfloor$ (if $j$ is even) or $\lceil t/2\rceil$ (if $j$ is odd) have the other parity; the four cases give the values stated. The reflection $j\mapsto 2k-j$ preserves types and reverses every transition, so $g_j=f_{2k-j}$. For $k$ even the minimum $a$ of $f$ is attained at $D$ levels (odd $j\le k$), so $\min\alpha=a\Lambda+am$; the minimum of $g$ is attained at $H$ levels, where friendly incoming sets are lines: $\min\beta=a\Lambda+am$. For $k$ odd, $f_j=a$ only at $H$ levels and every $D$ level has $f_j=a+1$, whose $\alpha_j=(a+1)\Lambda+am\ge aP+(a+1)m$ by $(*)$; so $\min\alpha=aP+(a+1)m$; $g_j=a$ only at $H$ levels, where $\beta_j=a\Lambda+(a+1)m$. □

So, with friendly point sets of size $P$, friendly line sets of size $\Lambda$ and non-friendly sets of size $m$, the alternating layout maximizes $K_{\mathcal{A}}+K_{\mathcal{B}}$ (and for $k$ even each of them) over all regular tournaments and typings. Its slot counts are

$$K_{\mathcal{B}}(q,k)=K_{\mathcal{A}}(q,k)=\tfrac{k}{2}(\Lambda+m)\ (k\text{ even}),\qquad K_{\mathcal{B}}=\tfrac{k-1}{2}P+\tfrac{k+1}{2}m,\quad K_{\mathcal{A}}=\tfrac{k-1}{2}\Lambda+\tfrac{k+1}{2}m\ (k\text{ odd}).$$

For $k=1$ this is $m$ on both sides, `D7_HEXAGON.md` Corollary 9.1.

*The layouts of the builds* [Computer-checked against the builders' logs]. The free slots per level, in level order:

| $k$ | types | $\alpha_j$ | $\beta_j$ |
|---|---|---|---|
| 2 | $HDHDH$ (alternating) | $P{+}m$, $\Lambda{+}m$, $P{+}m$, $2\Lambda$, $P{+}m$ | $\Lambda{+}m$, $2P$, $\Lambda{+}m$, $P{+}m$, $\Lambda{+}m$ |
| 3 | $HHDHHDD$ | $P{+}2m$, $P{+}2m$, $2\Lambda{+}m$, $2P{+}m$, $2P{+}m$, $2\Lambda{+}m$, $2\Lambda{+}m$ | $2\Lambda{+}m$, $2\Lambda{+}m$, $2P{+}m$, $\Lambda{+}2m$, $\Lambda{+}2m$, $2P{+}m$, $2P{+}m$ |
| 4 | $HHDHDHDHD$ (alternating, rotated) | $2P{+}2m$, $2P{+}2m$, $2\Lambda{+}2m$, $2P{+}2m$, $2\Lambda{+}2m$, $2P{+}2m$, $3\Lambda{+}m$, $2P{+}2m$, $3\Lambda{+}m$ | $2\Lambda{+}2m$, $2\Lambda{+}2m$, $3P{+}m$, $2\Lambda{+}2m$, $3P{+}m$, $2\Lambda{+}2m$, $2P{+}2m$, $2\Lambda{+}2m$, $2P{+}2m$ |

The $k=3$ layout of the builds is not the alternating one but has the same minima. With the values of each build these vectors equal, entry by entry, the vectors printed by `typed_build.py` (`d7thm/cells.out`).

*Computer-checked.* `d7thm/slots.py`: for $k\le 8$ every typing of the rotational tournament (the optimum is the one of Theorem 2.2, numerically for $q=11$ with $m=768$, $q=13$ with $m=792$, $q=16,101,1009$); every labelled regular tournament on 5 and 7 levels (24 and 2,640) with every typing; SAT (CaDiCaL) on $9,11,13$ levels finds no regular tournament and typing with $f_j,g_j\ge a$ everywhere, $f_j\ge a+1$ on $D$ levels and $g_j\ge a+1$ on $H$ levels (one combined instance; (b) and (c) themselves are proved above). Independently squad card c6: all regular tournaments on $3,5,7$ levels, 200,000 random ones on 9 and 11 levels, annealing, and the closed forms of Theorem 2.2 for $k\le 40$. Card c6 also found that without $(*)$ the value of $\min\alpha$ for odd $k$ changes (the $D$ levels become the minimum); this happens only for $k$ far above $q$ (for instance $k\ge 27$ at $q=11$), outside the range of the bound.

# 3. The bound

**Theorem 3.1** [new; Theorem]. Let $q$ be a prime power, $q$ odd or $q\ge 4$, let $1\le k\le q-1$, and let $K_{\mathcal{B}}(q,k)$, $K_{\mathcal{A}}(q,k)$ be the slot counts of Theorem 2.2 with $P=P(q)$, $\Lambda=\Lambda(q)$ and $m=q^2+1$. If an $\mathrm{HG}(K_1,K_2;q-k)$ exists with $K_1\le K_{\mathcal{B}}$ and $K_2\le K_{\mathcal{A}}$, then

$$n_b(q+1+k,\,7)\ \ge\ 2(2k+1)N(q)+K_1+K_2.$$

*Proof.* Take $2k+1$ copies of the incidence graph of $H(q)$, the alternating layout of Theorem 2.2, the identity on friendly transitions with freed sets $R_H=\mathcal{P}_S$ (source $H$) and $R_L$ (source $D$) of Corollary 1.8, and on non-friendly transitions the local duality $\sigma$ ($H\to H$) or $\sigma^{-1}$ ($D\to D$) of Proposition 1.9 with the $q^2+1$ freed elements. Every

friendly transition is repaired by Theorem B, every other one by Theorem C; $(*)$ holds since $k \le q-1$ (§2), so Theorem 2.2 gives the slot counts. Attach $K_1$ hubs of colour $\mathcal{B}$ and $K_2$ of colour $\mathcal{A}$ to free slots in any way (one slot per level each; unused slots stay free) and use the given graph as hub graph. Theorem 0.1 applies. □

The same holds with $m$ replaced by the size of any non-friendly set satisfying Theorem C, provided $(*)$ holds; and for any generalized hexagon of order $(q,q)$ with a spread, an exact whisker set with the property of Theorem 1.7, and a partial ovoid of a size $m$ satisfying $(*)$ (for instance $m \le q^2+1$), by the same proof.

**Corollary 3.2** [new; Theorem when the hub graph exists]. With full slots ($K_1 = K_{\mathcal{B}}$, $K_2 = K_{\mathcal{A}}$) Theorem 3.1 gives $n_b(q+1+k,7) \ge F_k(q)$, where

$$F_k(q) = 2(2k+1)(q+1)(q^4+q^2+1) + \begin{cases} k\,(q^4+2q^2+q+1), & k \text{ even}, \\ \frac{k-1}{2}\,(2q^4+q^3+q^2+2q+1) + (k+1)(q^2+1), & k \text{ odd}. \end{cases}$$

In particular

$$F_2(q) = 10N(q)+2q^4+4q^2+2q+2, \quad F_3(q) = 14N(q)+2q^4+q^3+5q^2+2q+5, \quad F_4(q) = 18N(q)+4q^4+8q^2+4q+4.$$

The gain over the level-graph value $2(2k+1)N(q)$ (the 2013 table value when $q+1+k \le 16$) is $kq^4+O(q^2)$ for $k$ even and $(k-1)q^4+\frac{k-1}{2}q^3+O(q^2)$ for $k$ odd; the earlier general statement, `D7_HEXAGON.md` Corollary 9.3 ($k=2$), gave $2(q+2)(q^2+1) \approx 2q^3$, and none was stated there for $k \ge 3$ except for $q = 3^h$ (Corollary 9.2). With a larger non-friendly set $m$ the formula changes by $k(m-q^2-1)$ for $k$ even and $(k+1)(m-q^2-1)$ for $k$ odd.

*Computer-checked* (`d7thm/cells.py`). The closed forms agree with the slot formulas for $q \le 37$, $k \le 6$.

# 4. Hub graphs

The bound needs $\mathrm{HG}(K_{\mathcal{B}}, K_{\mathcal{A}}; q-k)$ with $K \approx \lfloor k/2 \rfloor q^4$ hubs per side and degree $q-k$. The Moore-type bound for same-colour distance 6 is about $(q-k)^6$, so there is room, but the explicit hub graph of `D7_HEXAGON.md` (Lemma 7.1, a plane with pendant trees) has only about $\frac{4}{27}(q-k)^4$ vertices per colour. Write $K_0(s) = (s+1)(s^4+s^2+1)$ for the number of points of a hexagon of order $(s,s)$.

**Lemma 4.1** (*blown-up hexagon*) [Theorem; elementary]. Let $s$ be a prime power and $c \ge 1$. Replace every point and every line of $H(s)$ by between 1 and $c$ copies, and join two copies when the originals are incident. The result is an $\mathrm{HG}(K_1, K_2; c(s+1))$ for any chosen $K_1, K_2 \in [K_0(s), cK_0(s)]$, and any two of its vertices of different colours are at distance at most 5.

*Proof.* A path between copies of $x \ne y$ projects to a walk from $x$ to $y$ of the same length, and a geodesic from $x$ to $y$ lifts to copies, so the distance is $d(x,y)$; two copies of one element are at distance 2. A copy of $x$ has degree $\sum_{y \text{ I } x} t_y \le c(s+1)$. □

**Lemma 4.2** (*sums*) [new; Theorem]. Let $Y_1, \dots, Y_r$ be bipartite graphs with colour classes $(A_i, B_i)$, each with all same-colour distances at most 6, and for $i < r$ also all different-colour distances at most 5. Suppose $|A_j| \le |B_i|$ and $|B_j| \le |A_i|$ for $i < j$. For every $i < j$ add an injective map $A_j \to B_i$ and an injective map $B_j \to A_i$, as edges. The result is an $\mathrm{HG}(\sum |A_i|, \sum |B_i|;\ \max_i \Delta(Y_i) + r - 1)$.

*Proof.* The new edges join the colours $A$ and $B$, so the graph is bipartite with classes $\bigcup A_i$, $\bigcup B_i$. A vertex of $Y_i$ gains one edge to each $Y_j$, $j < i$, and at most one from each $Y_j$, $j > i$. Same-colour pairs inside one $Y_i$ are at distance at most 6 in $Y_i$. For $x \in A_j$ and $z \in A_i$ with $i < j$, $x$ has a neighbour $b \in B_i$, and $d_{Y_i}(b,z) \le 5$ since $i < r$; so $d(x,z) \le 6$. The same for $B$. □

**Lemma 4.3** (`D7_HEXAGON.md`, Lemma 7.1, with the colours sized separately) [restatement; Theorem]. Let $s$ be a prime power, $n = s^2+s+1$, $t = \delta - s - 1 \ge 0$, $u = \delta - 1$. Attach to the incidence graph of $\mathrm{PG}(2,s)$ $c_P$ pendant vertices at the points and $c_L$ at the lines (at most $t$ per core vertex), and $g_P$, $g_L$ pendant vertices below these (at most $u$ per vertex). This is an $\mathrm{HG}(n+c_L+g_P,\ n+c_P+g_L;\ \delta)$. *Proof.* The distance argument of Lemma 7.1 does not use the numbers of pendant vertices. □ (Different-colour distances can be 7 here, so such a part can only be the last part $Y_r$ in Lemma 4.2; squad card c5 confirmed this with a negative control.)

**Theorem 4.4** [new; Theorem; (b) computer-assisted: the decompositions are found and verified by a program]. Let $k \ge 2$. An $\mathrm{HG}(K_{\mathcal{B}}(q,k), K_{\mathcal{A}}(q,k); q-k)$ (full slots of Theorem 2.2, $m = q^2+1$) exists

(a) for every prime power $q \geq 300k$;

(b) for $2 \leq k \leq 12$, for every prime power $q \geq q_k$, where

| $k$ | 2 | 3 | 4 | 5 | 6 | 7 | 8 | 9 | 10 | 11 | 12 |
|---|---|---|---|---|---|---|---|---|---|---|---|
| $q_k$ | 16 | 23 | 23 | 29 | 23 | 31 | 29 | 37 | 32 | 43 | 41 |

These are the thresholds of our search (`d7thm/hubopt.py`), not of the method; they are not monotone in $k$.

*Proof.* (a) Let $q \geq 300k$ (so $q \geq 600$). The slot counts satisfy $q^4 < \Lambda \leq K_{\mathcal{A}} \leq K_{\mathcal{B}}$, $K_{\mathcal{B}}/K_{\mathcal{A}} \leq P/\Lambda \leq 1 + 1/q$ and $K_{\mathcal{B}} \leq kP \leq k(q+1)^4$. Put $x = K_{\mathcal{A}}^{1/5} - 1$ ($> q^{4/5} - 1 > 160$). By Nagura's theorem [Na52] (for $y \geq 25$ there is a prime in $(y, \frac{6}{5}y)$) there is a prime $s$ with $x/1.2 < s < x$. Then $K_0(s) < K_0(x) < (x+1)^5 = K_{\mathcal{A}}$, and, with $K_0(y) \geq y^5$, $x^5 \geq K_{\mathcal{A}}(1 - 5K_{\mathcal{A}}^{-1/5})$ (Bernoulli) and $K_{\mathcal{A}}^{-1/5} < q^{-4/5} \leq 0.006$,

$$\frac{K_{\mathcal{B}}}{K_0(s)} < \frac{1.2^5 K_{\mathcal{B}}}{x^5} \leq \frac{2.4884\,(1 + 1/q)}{1 - 0.03} < 3.$$

So Lemma 4.1 with $c = 3$ gives an $\mathrm{HG}(K_{\mathcal{B}}, K_{\mathcal{A}}; 3(s+1))$, and $3(s+1) < 3(x+1) = 3K_{\mathcal{A}}^{1/5} \leq 3k^{1/5}(q+1)^{4/5} = 3(q+1)(k/(q+1))^{1/5} \leq 3 \cdot 300^{-1/5}(q+1) < 0.959\,(q+1) \leq q - k$, the last step because $0.041(q+1) \geq 12.3k \geq k+1$. (b) For $q \leq 200$ the program `d7thm/hubopt.py` (an enumeration of multisets of parts with bounds) writes for each $(q, k)$ a decomposition into parts of Lemmas 4.1–4.3, combined by Lemma 4.2, and verifies it (sizes add up, the order condition, the degrees, the pendant-tree parameters, which `hubopt.out` prints); each line can be checked by hand. Examples: $(q, k) = (16, 2)$, $K = 66{,}065$: $H(8) + H(7) + H(5)$ and two copies of $H(4)$ blown up to 2,730 and 2,372 (degrees $9+4$, $8+4$, $6+4$, $10+4 \leq 14$); another one, built and checked below, is $H(8) + H(7) + H(5) + H(5)$ and a pendant-tree part on $\mathrm{PG}(2, 5)$ with 124 and 116 vertices at depth 1 and 1,049 and 1,041 at depth 2 (1,196 per colour, degree 10). $(q, k) = (23, 3)$: $H(11) + H(9) + H(8)$, the last blown up to 49,506 and 37,867. Below the full slots the same search gives the hub graphs of §5.2, for instance $(13, 4)$: $H(7) + H(7)$, two hexagons of order 7 joined by a perfect matching from the points of each to the lines of the other (39,216 per colour, degree $8 + 1 = 9$). For $200 < q \leq 2 \cdot 10^5$ Lemma 4.1 alone suffices (`d7thm/hubcap.py`, which finds $s$ and $c$ for every such $q$); for $q > 2 \cdot 10^5 \geq 300k$ use (a). □

*Computer-checked.* `d7thm/build_cert.py` builds the hub graph of a certificate (hexagons from `research/d10D7/hexgen2.c`, random injections) and `hgcheck edges` checks it: $(q, k) = (16, 2)$, $\mathrm{HG}(66065, 66065; 14)$ (the one behind $(19, 7)$), $(17, 2)$, $\mathrm{HG}(84117, 84117; 15)$, and $(23, 3)$, $\mathrm{HG}(293092, 281453; 20)$: no same-colour pair at distance more than 6, degrees as claimed (`d7thm/cert_checks.out`).

**Computer-checked hub graphs below** $q_k$ (`d7thm/hgcheck.c`, own code: bit-parallel breadth-first search from every hub; `d7thm/hgcheck_lab.out`, `d7thm/hgrun.out`). The lab's stored hub graphs, read from the solution files, re-checked standalone: $\mathrm{HG}(15541, 15541; 9)$ [(14,7)], $\mathrm{HG}(29535, 29535; 11)$ [(16,7)], $\mathrm{HG}(32356, 30327; 10)$ and $\mathrm{HG}(32356, 30158; 10)$ [(17,7)], $\mathrm{HG}(25308, 25308; 9)$ [(18,7), PSL(2,37)], $\mathrm{HG}(66065, 66065; 14)$ [(19,7)], $\mathrm{HG}(70163, 66322; 13)$ [(20,7)], and the Haar graph over $\mathrm{PSL}(2, 29)$ of job "conics", $\mathrm{HG}(12180, 12180; 8)$ [(15,7), `conics/d15D7/hub_psl29.txt`]: all VALID. The two cluster-assisted ones [(15,7) 16,152, (18,7) 34,440] are not hub graphs in this sense (their same-colour pairs are joined through the levels; in the hub graph alone 452,256 and 654,360 ordered same-colour pairs are at distance more than 6, the second number as in `conics2/README.md`); for them (HG) is covered by the all-pairs checks of the whole graphs. New random graphs (`hgcheck rand`, up to four seeds, aimed 2-switch repair when $K \leq 60{,}000$) at the slot counts of Theorem 2.2 with $m = q^2 + 1$, for the $(q, k)$ with $11 \leq q < q_k$, $k \leq 5$ (ordered same-colour pairs at distance more than 6 are "far pairs"; Haar graphs of $\mathbb{Z}_K$ were also tried and never covered):

| $(q, k)$ | $\delta = q - k$ | $K_{\mathcal{B}}$ | $K_{\mathcal{A}}$ | $M(\delta)/K_{\mathcal{B}}$ | result |
|---|---|---|---|---|---|
| $(11, 2)$ | 9 | 14,895 | 14,895 | 20.1 | found (rand seed 2) |
| $(13, 2)$ | 11 | 28,913 | 28,913 | 38.4 | found (rand seed 1) |

| $(q,k)$ | $\delta = q-k$ | $K_{\mathcal{B}}$ | $K_{\mathcal{A}}$ | $M(\delta)/K_{\mathcal{B}}$ | result |
|---|---|---|---|---|---|
| $(11,3)$ | 8 | 16,228 | 15,017 | 8.5 | not found (last graph tried: 602,324 far pairs) |
| $(13,3)$ | 10 | 31,112 | 29,083 | 19.2 | not found (last graph tried: 44 far pairs) |
| $(16,3)$ | 13 | 70,163 | 66,322 | 46.4 | found (rand seed 1) |
| $(17,3)$ | 14 | 89,032 | 84,407 | 58.7 | found (rand seed 1) |
| $(19,3)$ | 16 | 137,924 | 131,425 | 88.5 | found (rand seed 1) |
| $(11,4)$ | 7 | 29,790 | 29,790 | 1.9 | not found (last graph tried: 270,820,464 far pairs) |
| $(13,4)$ | 9 | 57,826 | 57,826 | 5.2 | not found (last graph tried: 37,527,036 far pairs) |
| $(16,4)$ | 12 | 132,130 | 132,130 | 14.7 | not found (last graph tried: 13,824 far pairs) |
| $(17,4)$ | 13 | 168,234 | 168,234 | 19.4 | not found (last graph tried: 212 far pairs) |
| $(19,4)$ | 15 | 262,126 | 262,126 | 30.9 | found (rand seed 1) |
| $(11,5)$ | 6 | 32,334 | 29,912 | 0.6 | not found (last graph tried: 1,190,378,570 far pairs) |
| $(13,5)$ | 8 | 62,054 | 57,996 | 2.2 | not found (last graph tried: 1,152,049,654 far pairs) |
| $(16,5)$ | 11 | 140,069 | 132,387 | 7.9 | not found (last graph tried: 43,492,232 far pairs) |
| $(17,5)$ | 12 | 177,774 | 168,524 | 11.0 | not found (last graph tried: 5,026,754 far pairs) |
| $(19,5)$ | 14 | 275,486 | 262,488 | 19.0 | not found (last graph tried: 11,012 far pairs) |

We did not run $k = 5$ with $q \geq 23$ (more than 300,000 hubs per colour) or $k = 6$.

**Heuristic.** Let $M(\delta) = 1 + \delta(\delta-1) + \delta(\delta-1)^3 + \delta(\delta-1)^5$, the most vertices of one colour within distance 6 of a vertex in a graph of maximum degree $\delta$. In the table above a random hub graph worked exactly when $M(\delta)/K_{\mathcal{B}} \geq 20$; at 19.0–19.4 ($(19,5)$, $(13,3)$, $(17,4)$) tens to thousands of far pairs were left, and below that random graphs are far off. Where they fail the lab used Haar graphs over $\mathrm{PSL}(2,p)$ and paths through the levels, which reach fewer hubs than the slots (16,152 of 16,309 at $(15,7)$, 34,440 of 59,070 at $(18,7)$).

# 5. The cells

## 5.1 The fifteen built graphs

`d7thm/cells.py` recomputes, for every graph built in the jobs "conics" and "conics-2", the slot vectors of §2 from $q$, the type string, $|\mathcal{P}_S| = P(q)$, $|R_L|$ ($q^4+q$ without $S_0$, $q^4+q^2+q$ with it) and $m$, and the order $2(2k+1)N(q) + K_{\mathcal{A}} + K_{\mathcal{B}}$ from the hub counts. **Computer-checked**: all fifteen agree with the builders (`cells.out`): every slot vector equals the one printed in the make logs on branch `claude/conics-2-diameter-7-ic0oat`, and every order equals the $n$ of the solution file. The current graphs:

| cell | $q$, types | $m$ | hubs per side | hub graph | $n$ | all-pairs check | the bound follows from |
|---|---|---|---|---|---|---|---|
| (14,7) | 11, $HDHDH$ | 768 | 15,541 + 15,541 | random | 1,802,642 | passed | Thm 3.1 + computer (set, hub graph) |
| (15,7) | 11, $HHDHHDD$ | 768 | 16,152 + 16,152 | Haar, clusters | 2,512,488 | passed | all-pairs check |
| (16,7) | 13, $HDHDH$ | 792 | 29,535 + 29,535 | random | 4,081,410 | passed | Thm 3.1 + computer (set, hub graph) |
| (17,7) | 13, $HHDHHDD$ | 792 | 32,356 + 30,327 | random | 5,693,959 | passed | Thm 3.1 + computer (set, hub graph) |
| (18,7) | 13, $HHDHDHDHD$ | 792 | 34,440 + 34,440 | Haar, clusters | 7,309,092 | passed | all-pairs check |
| (19,7) | 16, $HDHDH$ | 257 | 66,065 + 66,065 | random | 11,316,940 | 9/16 ranges | **Theorem** ($= F_2(16)$) |
| (20,7) | 16, $HHDHHDD$ | 257 | 70,163 + 66,322 | random | 15,795,219 | 9/24 ranges | Thm 3.1 + computer (hub graph) ($= F_3(16)$) |

All use $\Lambda = q^4+q^2+q$ (with $S_0$). Hub graphs: "random" means random bipartite of degree $q-k$ (checked standalone, §4); "Haar, clusters" means a Haar graph over $\mathbb{Z}_{673} \rtimes \mathbb{Z}_{24}$ resp. $\mathrm{PSL}(2,41)$ whose same-colour pairs are partly joined through the levels (`conics2/README.md` §5). The all-pairs states are those of `conics2/README.md` at 11:44 UTC.

At $(19,7)$ and $(20,7)$ the built graphs use exactly the sets of Theorem 3.1 ($m = q^2+1$, the elliptic section), so their orders are $F_2(16)$ and $F_3(16)$. For $(19,7)$ every ingredient is proved: the sets (Corollary 1.8, Proposition 1.9), the repair of every transition (Theorems B and C), the slot counts (Theorem 2.2) and an $\mathrm{HG}(66065, 66065; 14)$ (Theorem 4.4: $H(8)+H(7)+H(5)+H(5)$ and a pendant-tree part, joined by Lemma 4.2). So

$$n_b(19,7) \;\geq\; 11{,}316{,}940$$

holds by proof, independently of the all-pairs check of the built graph (which uses a random hub graph and was not complete when job "conics-2" ended). For $(20,7)$ the hub graph $\mathrm{HG}(70163, 66322; 13)$ is not covered by Theorem 4.4; it is computer-checked (the builder's graph, re-checked standalone by `hgcheck`, and a second random graph, seed 1, `hgrun.out`); with the proved equal-sided hub graph of 65,737 hubs per colour the bound is 15,790,208. This is a statement about the theorems, not a record claim: the lab records values after the coordinator's own check.

**A larger (18,7) graph** [new; Theorem, built and partly computer-checked]. At $(q,k) = (13,4)$ the cluster-assisted Haar hub graph of the current build has 34,440 hubs per colour, while Lemma 4.2 gives

HG(39216, 39216; 9) ($H(7) + H(7)$, §4). With the build's sets (slots 59,070 per colour) or those of Theorem 3.1 (slots 57,826), Theorem 3.1 gives

$$n_b(18, 7) \;\geq\; 18 \cdot 402{,}234 + 2 \cdot 39{,}216 \;=\; 7{,}318{,}644,$$

9,552 more than the lab's 7,309,092. We built it with `conics2/typed_build.py` (same hexagon file, types $HHDHDHDHD$, the 792-point mirrored set, $S_0$ included; the hub graph from `d7thm/build_cert.py`): the builder re-checks every hypothesis (segment and dual segment sets, the local duality at all 792 points, same-colour hub distances at most 6, degrees 15 to 18) and writes $n$ = 7,318,644, 65,612,918 edges, edge-list sha256 `c80d9c0e6a569508b2314c3c3cd6c53babce0ecb844017fdf7f86f297426ed1a`; the solution file `research/frontier/sols/d18D7_hex13_typedwhisk_h7h7_n7318644.sol` rebuilds it byte for byte (`typed_build.py build`). **Computer-checked:** (i) its level part (all edges between level vertices, 64,554,086 of them) is identical to that of the lab's 7,309,092 graph, rebuilt from its solution file (sha256 `314d5db9…`, as recorded); (ii) `apsp_fast` (`conics/apsp_fast.c`) from all 78,432 hubs: bipartite, degrees 15 to 18, every hub within distance 7 of every vertex (`RESULT VALID d=18 D=7 n=7318644` for the sources 7,240,212 to 7,318,643); (iii) `apsp_fast` from three ranges of level vertices (sources 0–262,143 in level 0, 804,468–1,066,611 in level 1, 6,978,068–7,240,211 in level 8; 786,432 sources, 10.9% of the level vertices): all `RESULT VALID`, eccentricity at most 7. Pairs of level vertices are within distance 7 inside two levels by Theorems B and C, whose hypotheses the builder checked; a full all-pairs check (about 20 hours on 4 cores here) was not run and is left to the coordinator (`d7thm/d18D7/`).

## 5.2 The cells $\Delta = 14, \ldots, 26$

For each $\Delta$, the best $(q, k)$ with $q + 1 + k = \Delta$; "level graphs" is $\max 2(2k+1)N(q)$, the 2013 table value for $\Delta \leq 16$ and the lab's beyond-table value for $\Delta = 17, \ldots, 20$ (`research/frontier/beyond/BEYOND_TABLE.md`, branch `claude/new-cells-beyond-etos3t`). $F_k(q)$ is Corollary 3.2 with $m = q^2 + 1$; its hub graph is "proof" (Theorem 4.4), "computer" (`hgcheck`) or "open". "Proved" is the value of Theorem 3.1 with the hub graph that `d7thm/hubopt.py` finds below the slots (a sum by Lemma 4.2, shown in the last column; full slots where Theorem 4.4 applies). The search is not proved optimal, so these are values found, not the best the lemmas allow.

| cell | level graphs | $(q, k)$ | $F_k(q)$ | hub graph at full slots | proved | at $(q, k)$: hubs | hub graph of the proved value |
|---|---|---|---|---|---|---|---|
| (14,7) | 1,771,560 | (11, 2) | 1,801,350 | computer | 1,797,726 | (11, 2): 13,083 + 13,083 | H(5) + H(5) + H(5) + H(4) |
| (15,7) | 2,480,184 | (11, 3) | 2,511,429 | open | 2,503,620 | (11, 3): 11,718 + 11,718 | H(5) + H(5) + H(5) |
| (16,7) | 4,022,340 | (13, 2) | 4,080,166 | computer | 4,079,910 | (13, 2): 28,785 + 28,785 | H(7) + H(5) + H(5) + H(4) |
| (17,7) | 5,631,276 | (13, 3) | 5,691,471 | open | 5,686,116 | (13, 3): 27,420 + 27,420 | H(7) + H(5) + H(5) |
| (18,7) | 7,240,212 | (13, 4) | 7,355,864 | open | 7,318,644 | (13, 4): 39,216 + 39,216 | H(7) + H(7) |
| (19,7) | 11,184,810 | (16, 2) | 11,316,940 | proof | 11,316,940 | (16, 2): 66,065 + 66,065 | H(8) + H(7) + H(5) + H(4)x2 + H(4)x2 |
| (20,7) | 15,658,734 | (16, 3) | 15,795,219 | computer | 15,791,202 | (16, 3): 66,234 + 66,234 | H(8) + H(7) + H(5) + H(5) + H(4) |
| (21,7) | 21,120,372 | (17, 3) | 21,293,811 | computer | 21,290,219 | (17, 3): 85,440 + 84,407 | H(8) + H(8) + H(5) + H(5) + H(4)x2 |
| (22,7) | 27,154,764 | (17, 4) | 27,491,232 | open | 27,476,636 | (17, 4): 160,936 + 160,936 | H(9) + H(8) + H(8) + H(7) |
| (23,7) | 36,591,240 | (19, 3) | 36,860,589 | computer | 36,857,870 | (19, 3): 135,205 + 131,425 | H(9) + H(8) + H(7) + H(5)x2 + H(5) |

| cell | level graphs | $(q,k)$ | $F_k(q)$ | hub graph at full slots | proved | at $(q,k)$: hubs | hub graph of the proved value |
|---|---|---|---|---|---|---|---|
| (24,7) | 47,045,880 | $(19,4)$ | 47,570,132 | computer | 47,566,386 | $(19,4)$: 260,253 + 260,253 | H(9) + H(9) + H(9) + H(8) + H(7) + H(5) |
| (25,7) | 57,500,520 | $(19,5)$ | 58,038,494 | open | 58,013,214 | $(19,5)$: 256,347 + 256,347 | H(9) + H(9) + H(9) + H(8) + H(7) |
| (26,7) | 67,955,160 | $(19,6)$ | 68,741,538 | open | 68,663,784 | $(19,6)$: 354,312 + 354,312 | H(11) + H(11) |

At $(15,7)$ the lab's Haar graph over PSL$(2,29)$ (`CONICS.md` §3.4; re-checked above: same-colour distances at most 6 in the hub graph alone) fits the slots of Theorem 2.2 with $m = q^2+1$ as well, so Theorem 3.1 with that computer-checked hub graph gives $14N(11) + 2 \cdot 12{,}180 = 2{,}504{,}544$.

At $(18,7)$ the proved value, 7,318,644, exceeds the lab's built 7,309,092 (§5.1). The proved values exceed the level-graph values at every cell $\Delta = 14, \dots, 26$. For $k \geq 3$ the lab's notes had no general statement except for $q = 3^h$ (`D7_HEXAGON.md`, Corollary 9.2; §9 there says none covers $(15,7)$). The values with computer-found ingredients (the lab's sets $m = 768, 792$ and hub graphs) are the built values of §5.1 and are larger at $q = 11, 13$.

## 6. Summary: what is new

| result | label | status |
|---|---|---|
| Theorem 1.4: lines meeting $v$, whiskers and the spread lines missed by them form a dual segment set, for any hexagon, spread and whisker set; maximality criterion | new in this form (special cases: `CONICS.md` Lemma 3.1, `conics2` Prop. 1(c),(d)) | Theorem, coordinate-free |
| Proposition 1.5: in a dual segment set whose lines meet only in spread points, at most $q^4+q$ lines are off the spread; $\mathcal{L}_v \cup W$ attains it | new | Theorem |
| Lemma 1.6: an exact whisker set misses at most $q^3-q^2$ spread lines, exactly $q^2$ under the "all or nothing" property | new | Theorem |
| Theorem 1.7, Corollary 1.8: $\lvert R_H\rvert = (q+1)(q^3+1)$, $\lvert R_L\rvert = q^4+q^2+q$, $R_L$ maximal, every odd $q$ and even $q \geq 4$ | restatement of `conics2` Prop. 1 with proof | Theorem; Computer-checked $q = 3, \dots, 16$ (squad c1–c3, builder; $q = 2$ checked to have no zero-free $G$) |
| Theorem 2.1: for every regular tournament and typing, the hub counts per side are at most $\lfloor k/2 \rfloor$ friendly sets, with the exact second-order term | new | Theorem; Computer-checked (all tournaments on $\leq 7$ levels, SAT to 13 levels, squad c6) |
| Theorem 2.2: the alternating layout attains Theorem 2.1 for every $k$; closed forms | new | Theorem; Computer-checked $k \leq 40$ |

| result | label | status |
|---|---|---|
| Theorem 3.1, Corollary 3.2: $n_b(q+1+k,7) \geq F_k(q)$ given the hub graph | new (for $k=2$ it improves `D7_HEXAGON.md` Cor. 9.3 from $\approx 2q^3$ to $\approx 2q^4$ above the level graph; nothing was stated for $k \geq 3$, $q \neq 3^h$) | Theorem |
| Lemma 4.2: sums of hub graphs; Lemma 4.1: blow-ups; Lemma 4.3: `D7_HEXAGON.md` Lemma 7.1 with the colours sized separately | 4.2 new; 4.1 elementary; 4.3 restatement | Theorem; Computer-checked (squad c5) |
| Theorem 4.4: the hub graphs at full slots exist for $q \geq q_k$ ($q_2 = 16$, $q_3 = q_4 = 23$, …, $q_k \leq 300k$) | new | Theorem; (b) computer-assisted (certificates by program for $q \leq 200$) |
| $n_b(19,7) \geq 11{,}316{,}940$ and $n_b(18,7) \geq 7{,}318{,}644$ (above the lab's 7,309,092) by proof; proved values above the level graphs at $(14,7)$–$(26,7)$ (§5.2) | new consequence | Theorem; the $(18,7)$ graph built, checked from every hub, full all-pairs check open |
| The formulas reproduce all fifteen built graphs; the lab's hub graphs re-checked standalone; new random hub graphs for $q < q_k$ | — | Computer-checked |
| Random hub graphs fail at $(q,k) = (11,3), (13,4)$; larger dual segment sets than $q^4+q^2+q$ for $q \geq 5$; a coordinate-free description of the $q^2$ lines with $\nu = 0$ | — | open; Heuristic remarks in §1.3 and §4 |

The theorems explain the values built by the two jobs (at $q = 16$ exactly) and give the same kind of bound for every $q$ and every $k$, with an explicit hub graph wherever $q \geq q_k$. One value is raised: at $(18,7)$ the explicit hub graph beats the computer-found one, giving 7,318,644 (a candidate until the coordinator's all-pairs check).

*Review.* A separate referee session re-derived the numerical steps (the Nagura chain, the closed forms, every table entry, all certificates of `hubopt.out` with its own checker, and Theorem 2.1 on all regular tournaments with at most 7 levels) and found no error in the proofs; it found that the first certificate search was far from optimal (now `hubopt.py`, which produced the $(18,7)$ hub graph) and several wording problems, all corrected here.

# Appendix: files

In `research/synthetic/d7thm/` (Python 3; `pip install python-sat` for `slots.py`; C with OpenMP for `hgcheck.c`; commands in `d7thm/README.md`):

- `slots.py` → `slots.out`: §2 (all typings of the rotational tournament for $k \leq 8$, all regular tournaments on 5 and 7 levels, SAT for 9–13 levels).
- `hubcap_lib.py`, `hubcap.py` → `hubcap.out`: Lemma 4.1 coverage for $q \leq 2 \cdot 10^5$ and the inequalities of Theorem 4.4(a); `hubopt.py` → `hubopt.out`: the certificates of Theorem 4.4(b) and of §5.2 for $q \leq 200$ (one line per $(q,k)$, pendant-tree parameters included); `build_cert.py`: the hub graph of a certificate as an edge list.
- `d18D7/`: the $(18,7)$ graph of §5.1 (builder log, level-part comparison, the check from every hub); its solution file is `research/frontier/sols/d18D7_hex13_typedwhisk_h7h7_n7318644.sol`.
- `hgcheck.c`: standalone check of a hub graph (same-colour distances by bit-parallel BFS from every vertex); modes `sol` (read a solution file), `rand` (random bipartite graph, optional aimed 2-switch repair), `haar` (Haar graphs of $\mathbb{Z}_K$). `hgcheck_lab.out`: the lab's hub graphs; `hgrun.py` → `hgrun.out`: random hub graphs at the slot counts of Theorem 2.2 for $q < q_k$.

- `cells.py` → `cells.out`: §5.1 and the closed forms of Corollary 3.2; `table.py` → `table.out`: §5.2.

Squad round `research/squads/d7thm-r1/`: `BRIEF.md`, cards `c1`–`c6` (code and logs), `scoreboard.jsonl`; logged in `research/squads/LOG.md`. The PDF `D7_THEOREMS.pdf` is made from this file by pandoc and tectonic (`d7thm/make_pdf.sh`).

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